

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)

11
priors compared

31,623
held-out plate appearances

3.05%
held-out HR rate

0.6020
best AUC · KNN comps

served prior

Held-out scores

Fitted on plate appearances before 2026-08-29, scored on the 31,623 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5862	1.61×	0.02950	0.13544	0.00535
KNN comps Prior is the pooled rate of the nearest comparables.	0.6020	1.55×	0.02945	0.13442	0.00478
SVR Support-vector regression on the same feature matrix.	0.5963	1.56×	0.02949	0.13515	0.00527
Random forest Forest regression; a learned similarity metric.	0.5956	1.48×	0.02948	0.13481	0.00643
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6020	1.55×	0.02945	0.13442	0.00478
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6020	1.55×	0.02945	0.13442	0.00478
Linear regression Ordinary least squares on the same features.	0.5963	1.51×	0.02947	0.13486	0.00416
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5974	1.55×	0.02949	0.13494	0.00556
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5964	1.53×	0.02948	0.13486	0.00505
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5962	1.55×	0.02949	0.13495	0.00593

BLEND WEIGHTS (FOREST VARIANT)

KNN 1.00 SVR 0.00 RF 0.00

SHRINKAGE K PER PRIOR

league 252 knn 252 svr 252 rf 252 core 252 ensemble 252
linear 252 logistic 252 nn 252 xgb 252

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,518 PA, actual HR rate 2.44%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5671	1.39×	0.02386	0.11499	0.01167
KNN comps Prior is the pooled rate of the nearest comparables.	0.5821	1.39×	0.02386	0.11479	0.01119
SVR Support-vector regression on the same feature matrix.	0.5696	1.28×	0.02403	0.11926	0.01804
Random forest Forest regression; a learned similarity metric.	0.5845	1.16×	0.02400	0.11611	0.01452
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5821	1.39×	0.02386	0.11479	0.01119
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5821	1.39×	0.02386	0.11479	0.01119
Linear regression Ordinary least squares on the same features.	0.5744	1.05×	0.02396	0.11712	0.01538
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5796	1.28×	0.02395	0.11633	0.01209
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5761	1.39×	0.02393	0.11629	0.01479
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5798	1.16×	0.02403	0.11727	0.01631

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Munetaka Murakami 17.0%, Miguel Vargas 14.6%, Randal Grichuk 14.1%, Jo Adell 13.1%, Kyle Teel 12.6%, Nathaniel Lowe 11.0%
Flat league	5/6	Munetaka Murakami 15.4%, Miguel Vargas 13.1%, Randal Grichuk 12.7%, Kyle Teel 12.4%, Jo Adell 12.4%, Chase DeLauter 10.9%
KNN comps	6/6	Munetaka Murakami 17.0%, Miguel Vargas 14.4%, Randal Grichuk 13.9%, Kyle Teel 13.1%, Jo Adell 12.9%, Nathaniel Lowe 11.0%
SVR	6/6	Munetaka Murakami 17.4%, Randal Grichuk 14.7%, Miguel Vargas 14.5%, Jo Adell 13.4%, Kyle Teel 12.0%, Nathaniel Lowe 10.8%
Random forest	6/6	Munetaka Murakami 17.2%, Miguel Vargas 14.2%, Randal Grichuk 13.7%, Jo Adell 13.1%, Kyle Teel 12.9%, Nathaniel Lowe 11.3%
Core (KNN+SVR)	6/6	Munetaka Murakami 17.0%, Miguel Vargas 14.4%, Randal Grichuk 13.9%, Kyle Teel 13.1%, Jo Adell 12.9%, Nathaniel Lowe 11.0%
Forest blend	6/6	Munetaka Murakami 17.0%, Miguel Vargas 14.4%, Randal Grichuk 13.9%, Kyle Teel 13.1%, Jo Adell 12.9%, Nathaniel Lowe 11.0%
Linear regression	5/6	Munetaka Murakami 16.8%, Miguel Vargas 14.5%, Randal Grichuk 13.8%, Jo Adell 13.0%, Kyle Teel 12.7%, Colson Montgomery 10.9%
Logistic regression	5/6	Munetaka Murakami 17.0%, Randal Grichuk 14.7%, Miguel Vargas 14.7%, Jo Adell 13.3%, Kyle Teel 11.8%, Colson Montgomery 11.4%
Neural net (torch)	6/6	Munetaka Murakami 17.0%, Miguel Vargas 14.5%, Randal Grichuk 14.5%, Jo Adell 13.4%, Kyle Teel 12.2%, Nathaniel Lowe 11.0%
XGBoost	6/6	Munetaka Murakami 17.3%, Miguel Vargas 14.5%, Randal Grichuk 14.0%, Jo Adell 13.2%, Kyle Teel 12.8%, Nathaniel Lowe 11.1%

Hitter by hitter

The top 40 bats by the served probability.

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (T
1Munetaka Murakami	slate17.0%	15.4%	17.0%	17.4%	17.2%	17.0%	17.0%	16.8%	17.0%	
2Miguel Vargas	CWS14.6%	13.1%	14.4%	14.5%	14.2%	14.4%	14.4%	14.5%	14.7%	
3Randal Grichuk	CWS14.1%	12.7%	13.9%	14.7%	13.7%	13.9%	13.9%	13.8%	14.7%	
4Jo Adell	LE13.1%	12.4%	12.9%	13.4%	13.1%	12.9%	12.9%	13.0%	13.3%	
5Kyle Teel	CWS12.6%	12.4%	13.1%	12.0%	12.9%	13.1%	13.1%	12.7%	11.8%	
6Nathaniel Lowe	CLE11.0%	10.7%	11.0%	10.8%	11.3%	11.0%	11.0%	10.9%	10.8%	
7Chase DeLauter	CLE10.8%	10.9%	10.7%	10.6%	10.7%	10.7%	10.7%	10.6%	10.5%	
8Angel Martínez	CLE10.8%	10.8%	10.6%	10.8%	10.7%	10.6%	10.6%	10.6%	10.6%	
9Colson Montgomery	CWS10.5%	9.5%	10.6%	10.1%	10.7%	10.6%	10.6%	10.9%	11.4%	
10Travis Bazzana	CLE10.1%	10.1%	9.9%	9.7%	10.1%	9.9%	9.9%	9.9%	9.7%	
11José Ramírez	CLE10.0%	10.0%	9.8%	9.6%	9.9%	9.8%	9.8%	10.1%	10.3%	
12Tristan Peters	CWS9.2%	9.8%	9.4%	9.6%	9.6%	9.4%	9.4%	9.5%	9.5%	
13Sam Antonacci	CWS8.5%	10.2%	8.8%	7.9%	8.9%	8.8%	8.8%	8.7%	8.5%	
14Chase Meidroth	CWS7.4%	8.6%	7.7%	8.1%	8.0%	7.7%	7.7%	7.8%	7.7%	
15Braden Montgomery	CWS7.0%	7.9%	6.7%	6.6%	6.7%	6.7%	6.7%	6.5%	6.4%	
16Bryan Rocchio	CLE6.8%	7.2%	6.6%	6.8%	6.4%	6.6%	6.6%	6.5%	6.5%	
17Austin Hedges	CLE6.1%	7.6%	6.7%	6.7%	6.7%	6.7%	6.7%	6.8%	6.5%	
18Steven Kwan	CLE4.7%	7.9%	5.6%	4.0%	4.0%	5.6%	5.6%	3.4%	4.8%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.